

Question Paper

Exam Date & Time: 29-Nov-2025 (02:00 PM - 05:00 PM)



MANIPAL ACADEMY OF HIGHER EDUCATION

MANIPAL INSTITUTE OF TECHNOLOGY - BENGALURU
SEVENTH SEMESTER B.TECH. DEGREE EXAMINATIONS - NOVEMBER 2025

Database and Application Security [IT_4060]

Marks: 50

Duration: 180 mins.

PART-A

Answer all the questions.

- 1) Analyse and differentiate between the following pairs of digital authentication methods based on their core operational principle and primary use case: (5)
- A)
- a) Kerberos and SSL
 - b) LDAP and RADIUS
 - c) Digital Certificate (PKI) and SRP (Secure Remote Password)
 - d) Digital Token and Digital Card
 - e) NTLM and any modern authentication method of your choice, justifying why it is considered obsolete.

Course Outcome: "CO1" / Cognitive Level: "Analysis"

- B) Apply your knowledge of effective password policy design to create a set of three specific, enforceable rules for a corporate environment. For each rule, briefly state its purpose in enhancing security. (3)

Course Outcome: "CO2" / Cognitive Level: "Application"

- C) Explain the primary security risk associated with using database links with a neat diagram, despite their utility in providing transparent data access. (2)

Course Outcome: "CO3" / Cognitive Level: "Application"

- 2) Analyse the two privilege revocation models described in the diagrams (ADMIN OPTION vs. GRANT OPTION) by answering the following: (5)

- A)
- a) Contrast the key difference in the cascading effect when a privilege is revoked from a user who granted it to others using the ADMIN OPTION versus the GRANT OPTION.
 - b) Evaluate which of these two models (ADMIN OPTION or GRANT OPTION) more strictly enforces the Principle of Least Privilege. Justify your answer.
 - c) Explain how the concept of Temporary Privilege Assignment could be used to mitigate a potential security risk inherent in the ADMIN OPTION model.

Course Outcome: "CO2" / Cognitive Level: "Application"

- B) A company is designing a secure web service for online banking. Apply the core concepts of a web services security blueprint by: (3)
- a) Naming two essential security components that must be part of the architecture.
 - b) Drawing a simple diagram showing how these components interact between the client and the web service to protect data.

Course Outcome: "CO3" / Cognitive Level: "Application"

- C) Applying your knowledge of how information systems are categorized, explain why a Transaction Processing System (TPS) is classified as supporting short-term, operational goals rather than long-term, strategic ones. (2)

Course Outcome: "CO1" / Cognitive Level: "Application"

- 3) Analyse why monitoring database links requires a different approach than monitoring regular user activity, and explain the role of the "Specialized User" method in overcoming this challenge. (5)

A)

Course Outcome: "CO3" / Cognitive Level: "Analysis"

- B) Case Study: Data Breach at "City Health Clinic" (3)

City Health Clinic recently digitized its patient records. A disgruntled former employee used old login credentials to access the system. They altered the dosage information in several patient prescriptions and exported a list of patients with sensitive health conditions, which was later found for sale online. Additionally, the clinic's server crashed due to a simple hardware failure, taking two days to restore from backups, during which time doctors could not access any patient histories.

Q. Applying the CIA Triad, analyse this security breach.

a) For each of the three CIA principles, identify a specific action from the case that violate it.

b) For one of the violations you identified, suggest a specific security measure that could have prevented it.

Course Outcome: "CO1" / Cognitive Level: "Application"

- C) The text emphasizes moving beyond pre-defined checklists. Apply this advice by explaining why a standardized security checklist would be insufficient for securing a distributed data environment that uses both database replication and mobile databases. (2)

Course Outcome: "CO3" / Cognitive Level: "Application"

- 4) Analyse the relationship between password hashing with salting and the effectiveness of a brute-force attack. Explain how the technical characteristics of a hash function, as mentioned in the presentation, directly disrupt the systematic methodology a brute-force attack relies on. (5)
- A)

Course Outcome: "CO4" / Cognitive Level: "Analysis"

- B) A developer needs to create a new web service operation that returns a list of active customers from a DB2 table. Using the WOF architecture, describe the specific purpose and role of the DADX file in this process. (3)

Course Outcome: "CO3" / Cognitive Level: "Application"

- C) Analyse the analogy "Bank Building - OS, Safe - DB, Money - Data" used to describe the OS security environment. Explain how this model illustrates the critical dependency of database security on the underlying operating system's integrity with a small diagram. (2)

Course Outcome: "CO1" / Cognitive Level: "Analysis"

- 5) Apply your understanding of fundamental security models by: (5)

a) Defining the Access Matrix Model and explaining its core components.

A)

b) Defining the Access Modes Model and explaining its core components.

c) Stating one key difference in how these two models structure access control information.

Course Outcome: "CO2" / Cognitive Level: "Application"

- B) A company is launching an internal cyber awareness campaign. Apply the provided definitions to explain why educating employees about 'attack vectors' like social engineering and suspicious email attachments is a direct method of safeguarding networks and computer systems from 'unauthorized digital access'. (3)

- C) Analyse the limitations of a common security measure (input validation) when confronted with a specific attack vector (DOM-Based XSS). (2)

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Event Name	Subject
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058-END SEMSTER-ODD-2025 IT_4060:Database and Application Security

#	Course Outcome	Description	Marks
1	1	CO 1	12
2	2	CO 2	13
3	3	CO 3	15
4	4	CO 4	10
Total			50

* **NA** - CO Not Selected

Event Name	Subject
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058-END SEMSTER-ODD-2025 IT_4060:Database and Application Security

#	Level	No of Questions	Marks (%)
1	Remember	0	0(0%)
2	Understand	0	0(0%)
3	Application	10	31(62%)
4	Analysis	5	19(38%)
5	Create	0	0(0%)
6	Evaluate	0	0(0%)
Total		15	50 (100%)